

Healthy Eating, Identities, and Design Opportunities in Social Media Spaces

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Abstract

Social media platforms like Instagram and TikTok play an increasingly influential role in shaping how people engage with food, health behaviors, and wellness—especially for those with dietary restrictions, such as teens and individuals with food allergies. While HCI research has explored health tracking, online communities, and food-related practices, there is still limited understanding of how users with specific health needs navigate algorithmic platforms over time to shape food practices, health goals, and identities. My dissertation addresses this gap by examining social media as both a site of engagement and a design material for supporting everyday healthy eating. Through this work, I contribute to HCI at the intersection of design, health, and social media.

CCS Concepts

• **Human-centered computing** → **Collaborative and Social Computing**; *Empirical studies in collaborative and social computing*.

Keywords

health; social media; food allergies; teenagers; healthy eating

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1 Introduction and Research Questions

Social media platforms have become influential spaces for sharing experiences, building communities, and navigating health challenges [1, 2, 6, 8]. My research examines how people interact with food content on these platforms and how such interactions shape everyday healthy eating. With the rise of food-related content [6, 8] and increasing public health concerns, I explore how social technologies influence dietary habits and support personal health and reflection.

Prior studies design systems to leverage social potential and affordances to encourage users to eat healthily. For example, allowing users to share and positively evaluate each other's meals can

casually encourage healthy eating habits [12]. Promoting healthy foods on young consumer social networks also has the potential to encourage them to adopt healthy lifestyles [9]. Also, existing work from personal informatics [5] shows that making tracking social could support people with healthy eating goals [2]. In design explorations, lightweight social tools were found to be effective in promoting reflection and mindfulness in healthy eating practices [4]. Ephemeral social media features could also be designed as personal informatics tools [16]. I hope to further explore opportunities where we could use social media as a design material [3, 7] to support healthy eating.

In my dissertation, I focus on people with different healthy eating needs and goals, particularly those with food allergies, and how they use social media to navigate their health and identity. These individuals face unique challenges as they engage with algorithm-driven platforms that may not fully cater to their specific needs. Their use of social media often reflects a balance between self-representation and managing their health-related goals, while also negotiating social norms and potential stigma. Drawing from Strauss's [11] concept of information work and Simpson's identity work [10], my research examines how individuals continually adapt their interactions, both online and offline, to align with their evolving health needs. By focusing on people with diverse dietary habits, goals, and needs, this work aims to highlight how social media can better support users with varied health goals and offer design insights for inclusive, health-focused platforms.

My goal is to explore the evolving landscape of social media design and its intricate relationship with human behaviors, and also examine how these platforms contribute to the cultivation of healthy lifestyles.

My research questions:

- RQ1: How do teens consume food content on social media, and how does that affect their healthy behaviors?
- RQ2: How do people with food restrictions navigate and use social media content in support of their health information and identity work?
- RQ3: How do individuals with food allergies use social media platforms, particularly TikTok, to express their health experiences and identities, and what norms and patterns emerge around food allergy discussions in these digital environments?
- RQ4: How can social media be used as a design material for technologies that support reflections and self-expressions of everyday healthy eating practices?

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2 Completed and Ongoing Work

My dissertation explores how social media platforms shape people's everyday healthy eating behaviors and identity work, particularly among teens and individuals managing food restrictions. Motivated by the growing influence of algorithmic feeds on food content consumption, I began by investigating how teenagers engage with food videos on TikTok through 15 interviews with teens across the U.S. This study revealed how teens use platform-curated content not only for food inspiration but also as a way to express identity, connect socially, and even negotiate body image. These insights demonstrated the profound effects that short-form food content—and the algorithms that serve it—can have on offline behaviors and long-term habits.

Building on this, I turned to individuals with food allergies, a population for whom food is not just personal but tied to risk and stigma. Through 18 semi-structured interviews with young adults with food allergies, I found that social media platforms are dual spaces of empowerment and emotional burden. Participants used social media to find community and support, but also developed careful strategies to manage their visibility and safety. This study emphasized the idea of “on-and-off identities,” where users shift between expressing and concealing aspects of themselves depending on the social and algorithmic context, around food and healthy eating.

To better understand the broader cultural and technological dynamics, I conducted a content analysis of 210 TikTok videos related to food allergies. This work investigates how platform norms shape the presentation of health experiences, the kinds of narratives that are amplified, and the affective labor required to share chronic illness online. Together, these findings point to the need for more inclusive, stigma-aware platform design that accounts for health-specific forms of expression and identity work.

These three studies set the foundation for my current and final project, which explores how social media can be used not just for immediate engagement, but as a long-term material for reflection and behavior change. This ongoing study involves interviews with users about how their food-related posting habits have shifted, and a design probe using Instagram's Story templates to prompt food-related reflection and communal sharing. The goal is to understand how ephemeral social media functions can support social and mindful engagement with food and health, and to develop design principles for platforms that better support long-term health practices, identity exploration, and social experiences.

In my ongoing study, I examine how existing social media features can be used as design materials to support lightweight, everyday reflection on food practices and identity work. Building on my earlier findings on algorithmic influence, visibility management, and evolving food narratives—particularly on platforms like TikTok—this study addresses a key gap: how tools for health reflection can be meaningfully embedded within the rhythms and affordances of everyday social media use.

Using Instagram's Story templates (such as “Add Yours”) as a design probe, I aim to investigate how users engage with lightweight, ephemeral, and socially embedded prompts over time, and how these interactions shape their self-expression, reflection, and sense of community.

3 Overall Contributions of My Dissertation Work

Building on Wobbrock et al.'s [17] framework of seven types of contributions to HCI, I created Table 1 to summarize the contributions of my dissertation work.

In sum, these contributions offer a holistic perspective on how social media can be thoughtfully designed to support healthy practices, identity formation, and inclusive engagement for users with varied health needs. Study 1 brings forward a theoretical model that explores how social media can encourage reflective, health-oriented behaviors over time. By examining algorithm-driven influences, Study 2 provides an in-depth understanding of how social media shapes identity work, particularly for users managing conditions like food allergies, revealing the complexities of agency and vulnerability in these spaces. Additionally, Study 3 uncovers social norms and expressions around food allergies on social platforms, offering key insights for designing digital environments that are inclusive and supportive. Finally, the designs from Study 4 leverages social media elements as a design material to promote social and sustained healthy eating, demonstrating the potential of social media to serve as a tool for long-term health support within HCI. Together, I aim to shape these studies into opinion contributions [17], offering a comprehensive framework for HCI researchers and designers working at the intersection of health, identity, and social media.

4 Next Steps

Future work will focus on piloting and analyzing the Instagram Stories design probe. I will begin by conducting a round of semi-structured interviews to understand participants' evolving engagement with food-related content, particularly how they reflect on their eating habits, health goals, and identity expression over time.

Following this, I will deploy a set of Instagram Story templates designed to prompt reflection around food practices, moods, and social interactions. In a combination of qualitative interview data and content analysis of story responses, I will examine how participants engage with these prompts and how they navigate visibility, audience boundaries, and emotional sharing in a social media context.

The analysis will focus on how social media affordances (e.g., ephemerality, social chains, limited feedback) influence health-related reflection and self-presentation. It will also investigate differences in engagement across users with varying health concerns, including those with food allergies.

Beyond this study, I aim to further explore how social media can be used not only for short-term sharing but also as a sustainable design material for long-term health behavior support and identity work. Future work will involve iterating on probe designs and developing design principles for inclusive, socially embedded health technologies. I also plan to connect findings from this study back to my earlier work on algorithmic visibility and on/off identity management to inform the broader design implications of reflective social media features.

Table 1: Research Questions and Contributions to HCI Research

Study	Research Questions	Contribution	Contribution Type	Progress
1	How do teens consume food content on social media, and how does that affect their healthy behaviors?	Theoretical model of a temporal-based social media-inspired reflective healthy practices; Understanding the role of algorithm-driven social media in health behaviors	Empirical, Theoretical	Complete; presented at CSCW 24 [14] & IDC 2022 [13]
2	How do people with food restrictions navigate and use social media content in support of their health information and identity work?	Understanding identity work for individuals with food allergies; Understanding agency and vulnerability in algorithmic environments, and how social media algorithms influence users' visibility and identity work	Empirical	Complete; presented at CHI 25 [15]
3	What messages are crafted on TikTok about food allergies, and what norms and patterns emerge around food allergy discussions in these digital environments?	Examining food allergy norms and expressions on social media; Informing design implications for inclusive and supportive digital platforms	Empirical, Dataset	Complete; planned for submission
4	How can social media be used as a design material for technologies that support reflections and self-expressions of everyday healthy eating practices?	Prototype; social media as design material to support long term healthy eating	Empirical, Artifact	Ongoing

5 Doctoral Consortium

The CSCW Doctoral Consortium comes at a pivotal time in my PhD, as I move into the final year of my dissertation. My earlier studies explored how teens and individuals with food allergies use algorithmic social media for food-related reflection, identity work, and navigating stigma. My current project builds on this by using Instagram's "Add Yours" feature as a design probe to investigate how social media can serve as a long-term material for health reflection.

I hope to receive feedback on refining my research probes, integrating empirical and design work, and framing my contributions to HCI and CSCW. I'm also eager to learn from others working on identity, health, and social technologies, and to explore future directions and collaborations that could strengthen the impact and coherence of my dissertation.

The Doctoral Consortium at CSCW 2025 offers an invaluable opportunity to gain insights and feedback from esteemed professors and talented peers. Given the interdisciplinary nature of my work, which intersects design, social media, and health behaviors, engaging with experts across diverse fields (e.g., Computer Science, AI, Psychology) and research areas (e.g., Science and Technology Studies (STS), marginalized user studies, mobile devices) would greatly enrich my research. I am also eager to explore collaborative opportunities with researchers working in these areas.

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